Dipayan **CHAKRABORTY**

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Technopôle Château-Gombert
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dipayan.chakraborty@centrale-med.fr / dipayancha@gmail.com
`https://dipayan5186.github.io/Website/`

Research domains: *identification problems in graphs, dominating sets, graph modification problems, graph algorithms, computational & parameterized complexity*

last updated : 13 mars 2026

Professional experience

Sep. 2025 - till date — **Assistant Professor (ATER in the French higher education system)**, *Centrale Méditerranée, Marseille – Laboratoire d'Informatique et Systèmes (LIS), Aix-Marseille University*, France

Feb. 2025 - Aug. 2025 — **Post-doctorate** (supervisor : **Faisal N. Abu-Khzam**), *Department of Computer Science and Mathematics, Lebanese American University*, Lebanon

Jan. 2022 - Janv. 2025 — **PhD scholar**, *Laboratoire d'Informatique, de Modélisation et d'Optimisation des Systèmes (LIMOS), Université Clermont Auvergne*, France

Director — **Annegret WAGLER**, *Professor, Université Clermont Auvergne*, France
Co-supervisor — **Florent FOUCAUD**, *Associate Professor, Université Clermont Auvergne*, France
Co-supervisor — **Michael HENNING**, *Professor, University of Johannesburg*, South Africa

Jan. 2023 - Jan. 2025 — **Reseach associate** (academic host : **Michael Henning**), *Department of Mathematics and Applied Mathematics, University of Johannesburg*, South Africa

Oct. 2022 - Nov. 2024 — **Part-time lecturer**, *Institut universitaire de technologie (IUT), Université Clermont Auvergne*, France

Jan. 2021 - Dec. 2021 — **Contractual researcher**, *as part of the project funded by SERB-SRG and titled "Complexity dichotomies of graph modification problems", Department of Computer Science and Engineering, Indian Institute of Technology Dharwad*, Karnataka, India

Aug. 2016 - Dec. 2020 — **Assistant Professor**, *Department of Basic Science & Humanities, Gargi Memorial Institute of Technology, Maulana Abul Kalam Azad University of Technology*, Baruipur, Kolkata, India

Jul. 2015 - Jul. 2016 — **Visiting lecturer**, *Department of Mathematics, Government College of Engineering and Ceramic Technology, Maulana Abul Kalam Azad University of Technology*, Kolkata, India

Education and Trainings

Jan. 2022 - Jan. 2025 — **PhD in Computer Science** (defended 9th December 2024)

Thesis title — **Structural and algorithmic aspects of identification problems in graphs**

Laboratoire d'Informatique, de Modélisation et d'Optimisation des Systèmes (LIMOS), Université Clermont Auvergne, France

PhD Jury

Director — **Annegret Wagler**, *Professor, Université Clermont Auvergne*, France
Co-supervisor — **Florent Foucaud**, *Associate professor, Université Clermont Auvergne*, France
Co-supervisor — **Michael Henning**, *Professor, University of Johannesburg*, South Africa

President & Examiner — **Paul Dorbec**, *Professor, Université de Caen-Normandie*, France
Examiner — **Ralf Klasing**, *Director of Research, CNRS, Université de Bordeaux*, France

Reviewer **Tero Laihonen**, *Professor, University of Turku*, Finlande
Reviewer **Arnaud Pêcher**, *Professor, Université de Bordeaux*, France

Sep. 2009 - Mar. 2012 **Master in mathematics** (ALGANT Erasmus Mundus)
Université de Bordeaux, France (1st year) – Università degli Studi di Padova, Italy (2nd year)

Master thesis defended March 2012 **Mordell-Weil theorems & the Birch and Swinnerton-Dyer conjecture**, *Università degli Studi di Padova*, Italy

Supervisor **Matteo Longo**, *Università degli Studi di Padova*, Italy

Aug. 2004 - June 2007 **Bachelor of Science** (Honours in mathematics)
St. Xavier's College, Kolkata, University of Calcutta, Kolkata, India

Publications and submissions in journals

Accepted Mar. 2026 **Tight (double) exponential bounds for identification problems : Locating-Dominating Set and Test Cover**, Dipayan Chakraborty, Florent Foucaud, Diptapriyo Majumdar and Prafullkumar Tale, *SIAM Journal on Discrete Mathematics* (2026)
`http://arxiv.org/abs/2402.08346`

Accepted Mar. 2026 **Identifying codes in triangle-free graphs of bounded maximum degree**, Dipayan Chakraborty, Florent Foucaud, Michael Henning and Tuomo Lehtilä, *Journal of Graph Theory* (2026)
`https://arxiv.org/abs/2403.17877`

Accepted Feb. 2026 **Partitioning the vertex set of a graph into a dominating set and a locating dominating set**, Dipayan Chakraborty, Florent Foucaud, Michael Henning and Tero Laihonen, *The Electronic Journal of Combinatorics* (2026)
`https://hal.science/hal-05022831`

Published June 2026 **Identifying open codes in trees and 4-cycle-free graphs of given maximum degree**, Dipayan Chakraborty, Florent Foucaud and Michael Henning, *Discrete Applied Mathematics* 386 : 319-333 (2026)
`https://doi.org/10.1016/j.dam.2026.02.032`

Published Feb. 2026 **Identifying codes in graphs of given maximum degree : Characterizing trees**, Dipayan Chakraborty, Florent Foucaud, Michael Henning and Tuomo Lehtilä, *Discrete Mathematics* 349(2) : 114826 (2026)
`https://doi.org/10.1016/j.disc.2025.114826`

Published Jan. 2026 **On full-separating sets and related codes in graphs**, Dipayan Chakraborty and Annegret K. Wagler, *Discrete Applied Mathematics* 379 : 551-569 (2026)
`https://doi.org/10.1016/j.dam.2025.10.001`

Published Nov. 2025 **On open-separating dominating sets in graphs**, Dipayan Chakraborty and Annegret K. Wagler, *Discrete Applied Mathematics* 375 : 215-238 (2025)
`https://doi.org/10.1016/j.dam.2025.05.045`

Published Jan. 2025 **A linear algorithm for radio k-coloring of powers of paths having small diameters**, Dipayan Chakraborty, Soumen Nandi, Sagnik Sen and Supraja DK, *Journal of Computer and System Sciences* 147 : 103577 (2025)
`https://doi.org/10.1016/j.jcss.2024.103577`

Published Dec. 2024 **Progress towards the two-thirds conjecture on locating-total-dominating sets**, Dipayan Chakraborty, Florent Foucaud, Anni Hakanen, Michael Henning and Annegret Wagler, *Discrete Mathematics* 347(12) : 114176 (2024)
`https://doi.org/10.1016/j.disc.2024.114176`

Published Dec. 2024 **On locating and neighbor-locating colorings of sparse graphs**, Dipayan Chakraborty, Florent Foucaud, Soumen Nandi, Sagnik Sen and Supraja DK, *Discrete Applied Mathematics* 358 : 366-381 (2024)
`https://doi.org/10.1016/j.dam.2024.07.012`

Published July 2024 **On three domination-based identification problems in block graphs**, Dipayan Chakraborty, Florent Foucaud, Aline Parreau and Annegret Wagler, *Fundamenta Informaticae (special issue : Iro Honkala's 60th birthday)*, 191(3-4) : 197-229 (2024)
https://doi.org/10.3233/FI-242179

Published Jan. 2023 **On clique numbers of colored mixed graphs**, Dipayan Chakraborty, Sandip Das, Soumen Nandi, Debdeep Roy and Sagnik Sen, *Discrete Applied Mathematics* 324 : 29-40 (2023)
https://doi.org/10.1016/j.dam.2022.08.013

Submitted Mar. 2025 **On the Structural Parameterizations of Locating-Dominating Set and Test Cover**, Dipayan Chakraborty, Florent Foucaud, Diptapriyo Majumdar and Prafullkumar Tale, *Manuscript* (2025)
(submitted to *Journal of Computer and System Sciences*)

Submitted Feb. 2025 **On lower bounds for cardinalities of several separating-dominating codes in graphs**, Dipayan Chakraborty and Annegret K. Wagler, *Manuscript* (2025)
(submitted to *Discrete Applied Mathematics*)

Submitted June 2024 **The $n/2$-bound for locating-dominating sets in subcubic graphs**, Dipayan Chakraborty, Anni Hakanen and Tuomo Lehtilä, *Manuscript* (2024)
(submitted to *Graphs and Combinatorics*)

Preprint 2026 **On the Complexity of Vertex-Splitting Into an Interval Graph**, Faisal N. Abu-Khzam, Dipayan Chakraborty, Lucan Isenmann and Nacim Oijid, *Manuscript* (2026)
https://doi.org/10.48550/arXiv.2602.04628

Publications and submisions in conferences with proceedings

Published Nov. 2025 **The interplay between domination and separation in graphs**, Dipayan Chakraborty and Annegret K. Wagler, *XIII Latin American Algorithms, Graphs, and Optimization Symposium (LAGOS 2025) Procedia Computer Science 273 : 78-85 (2025)*
https://doi.org/10.1016/j.procs.2025.10.283

Published June 2025 **Structural parameterization of locating-dominating set and test cover**, Dipayan Chakraborty, Florent Foucaud, Diptapriyo Majumdar and Prafullkumar Tale, *14th International Conference on Algorithms and Complexity (CIAC), Lecture Notes in Computer Science 15679 : 187-204 (2025)*
https://doi.org/10.1007/978-3-031-92932-8_13

Published Feb. 2025 **On full-separating sets in graphs**, Dipayan Chakraborty and Annegret Wagler, *11th International Conference on Algorithms and Discrete Applied Mathematics (CALDAM), Lecture Notes in Computer Science 15536 :73-84 (2025)*
https://doi.org/10.1007/978-3-031-83438-7_7

Published Dec. 2024 **Tight (double) exponential bounds for identification problems : Locating-Dominating Set and Test Cover**, Dipayan Chakraborty, Florent Foucaud, Diptapriyo Majumdar and Prafullkumar Tale, *35th International Symposium on Algorithms and Computation (ISAAC), Leibniz International Proceedings in Informatics 322 : 19 :1-19 :18 (2024)*
https://doi.org/10.4230/LIPIcs.ISAAC.2024.19

Published May 2024 **Open-separating dominating codes in graphs**, Dipayan Chakraborty and Annegret K. Wagler, *8th International Symposium on Combinatorial Optimization (ISCO), Lecture Notes in Computer Science 14594 : 137-151* (2024)
https://doi.org/10.1007/978-3-031-60924-4_11

Published Jan. 2024 **Location-Domination Type Problems Under the Mycielski Construction**, Silvia M. Bianchi, Dipayan Chakraborty, Yanina Lucarini and Annegret K. Wagler, *10th International Conference on Algorithms and Discrete Applied Mathematics (CALDAM), Lecture Notes in Computer Science 14508 : 255-269 (2024)*
https://doi.org/10.1007/978-3-031-52213-0_18

Published Sep. 2023 **Contracting edges to destroy a pattern : A complexity study**, Dipayan Chakraborty and Sandeep RB, *24th International Symposium on Fundamentals of Computation Theory (FCT), Lecture Notes in Computer Science 14292 : 118-131 (2023)*
https://doi.org/10.1007/978-3-031-43587-4_9

Published Sep. 2023 **Identifying codes in bipartite graphs of given maximum degree**, *with* Dipayan Chakraborty, Florent Foucaud, Tuomo Lehtilä, *XII Latin-American Algorithms, Graphs and Optimization Symposium (LAGOS), Procedia Computer Science 223 : 157-165 (2023)*
https://doi.org/10.1016/j.procs.2023.08.225

Published June 2023 **A linear algorithm for radio k-coloring powers of paths having small diameter**, Dipayan Chakraborty, Soumen Nandi, Sagnik Sen and DK Supraja, *34th International Workshop on Combinatorial Algorithms (IWOCA), Lecture Notes in Computer Science 13889 : 148-159* (2023)
https://doi.org/10.1007/978-3-031-34347-6_13

Published Jan. 2023 **On three domination-based identification problems in block graphs**, Dipayan Chakraborty, Florent Foucaud, Aline Parreau and Annegret Wagler, *9th International Conference on Algorithms and Discrete Applied Mathematics (CALDAM), Lecture Notes in Computer Science 13947 : 271-283 (2023)*
https://doi.org/10.1007/978-3-031-25211-2_21

Published Jan. 2023 **New bounds and constructions for neighbor-locating colorings of graphs**, Dipayan Chakraborty, Florent Foucaud, Soumen Nandi, Sagnik Sen and Supraja DK, *9th International Conference on Algorithms and Discrete Applied Mathematics (CALDAM), Lecture Notes in Computer Science 13947 : 121-133 (2023)*
https://doi.org/10.1007/978-3-031-25211-2_9

Conferences / Summer Schools

June 2025 **14th International Conference on Algorithms and Complexity**, (CIAC 2025)
Organised by : Libera Università Internazionale degli Studi Sociali Guido Carli, Italy

November 2024 **Journées Graphes et Algorithmes**, (JGA 2024)
Organised by : Université de Bourgogne, France

May 2024 **École de Printemps en Informatique Théorique**, (EPIT 2024)
Organised by : École Normale Supérieure de Lyon, France

February 2024 **10th International Conference on Algorithms and Discrete Applied Mathematics**, (CALDAM 2024)
Organised by : IIT Bhilai, Chhattisgarh, Inde

September 2023 **XII Latin-American Algorithms, Graphs and Optimization Symposium**, (LAGOS 2023)
Organised by : Instituto de Matemáticas, UNAM, Juriquilla, Mexique

June 2023 **CIMPA Research School on Graph Structure and Complex Network Analysis**, (CIMPA GSCN 2023)
Organised by : Nesin Mathematics Village, Sirinçe, Izmir, Turkie

February 2023 **9th International Conference on Algorithms and Discrete Applied Mathematics**, (CALDAM 2023)
Organised by : DAIICT, Gandhinagar, Inde

September 2022 **Maribor Graph Theory Conference**, (MGTC 2022)
Organised by : Faculty of Natural Sciences and Mathematics, University of Maribor, Slovenie

July 2022 **11th International Colloquium on Graph Theory and Combinatorics**, (ICGT 2022)
Organised by : LIRMM, Montpellier, France

june 2022 **School on Graph Theory**, *Murol, France*, (SGT 2022)
Organised by : LIMOS, Université Clermont Auvergne, France

May 2021 **Annual Meeting of Homomorphisms of Signed Graphs**, (HOSIGRA 2021)
Organised by : Université de Bordeaux, Université de Montpellier et Université de Paris Diderot, France

Reviews for journals and conferences

Conference **International Conference on Algorithms and Discrete Applied Mathematics**, *CALDAM*

International Workshop on Combinatorial Algorithms, *IWOCA*

Latin-American Algorithms, Graphs and Optimization Symposium, *LAGOS*

International Conference on Current Trends in Theory and Practice of Computer Science, *SOFSEM*

International Workshop on Graph-Theoretic Concepts in Computer Science, *WG*

International Computing and Combinatorics Conference, *COCOON*

Journal **Fundamenta Informaticae**

The Australasian Journal of Combinatorics

Research supervision

May 2025 - Dec. 2025 **Intern of 3rd year bachelor's degree, PSG College of Technology, Coimbatore, India**, *Subject of research : Identification problems in graphs*
Co-supervised with Prof. Florent Foucaud, Université Clermont Auvergne, France

May 2024 - July 2024 **Intern of 3rd year bachelor's degree, Chennai Mathematical Institute, Chennai, India**, *Subject of research : Local identification problems in graphs*
Co-supervised with Prof. Florent Foucaud, Université Clermont Auvergne, France

Organisation of scientific events

Organising committee (Mar. 2024) **41st International Symposium on Theoretical Aspects of Computer Science**, *Clermont-Ferrand, France*, (STACS 2024)
Organised by : Université Clermont-Auvergne, France

Teaching activities (in France)

Centrale Médietrranée, Château-Gombert, Marseille, France

Sep. 2025 - till date **Computer Science Engineering : Bachelor year 1** (2025-2026 ‖ 176 hours)

- **JAVA** (Tutorials + Practicals ‖ 48 hours)
  Basics of JAVA • Creating a class • Heritage & Polymorphisms • Interface • Exceptions • Reading and manipulating files
- **Algorithms** (Tutorials + Practicals ‖ 64 hours)
  Complexities of algorithms • Algorithms on trees • Programming of algorithms on trees • Algorithms on general graphs • Dynamic programming • Algorithms on texts • Texts and lexicographic trees
- **Database** (Tutorials + Practicals ‖ 40 hours)
  Functional dependencies and normalizations • Entity-relationship diagrams • Relational algebra • Execution of requests (SQL)

Nov. 2025 - Jan. 2026 **Computer Science Engineering : Bachelor year 2** (2025-2026 ‖ 18 hours)

- **Scientific Python** (Practicals ‖ 18 hours)
  Numpy • Matplotlib • Pandas • Seaborn • Scikit

Aix-Marseille Université, Saint-Jerôme, Marseille, France

Jan. 2025 - till date **Bachelor year 1 : Mathematics, Physics, Chemistry & Computer Science** (2025-2026 ‖ 30 hours)

- **Programming & algorithms** (Course + Tutorials + Practicals ‖ 30 hours)
  Proving an algorithm • Iterative and recursive algorithms & pseudo-codes • Testing of codes, developing a project, nd A particular project on percentages • Computational complexity of an algorithm • Complexity of a problem • Exponential Problems • Average-case complexity • Different types of sortings

Oct. 2025 - Dec. 2025 **Bachelor year 2 : Mathematics, Physics, Chemistry & Computer Science** (2025-2026 || 10 hours)

- **Graph theory and algorithms** (Practicals || 10 hours)

  Class of binairy trees • Huffman coding • Graphes • Computing running-times • Maximum Flow Problem (taxi of the Marne)

IUT Clermont Auvergne, *Université Clermont Auvergne*, Aubière, France

2022 - 2024 **Bachelor year 2 - Computer Science - Bachelor Universitaire de Technologie (BUT)** (2022-2024 || 42 hours)

- **Methods of optimisation** (Tutorials + Practicals || 42 hours)

  Introduction to linear programming • Integer linear programming • Utilising linear problem solver in Sagemath • Vertex cover algorithmes • Heuristic algorithmes for the Travelling Salesman

Teaching activities (in India)

Gargi Memorial Institute of Technology (GMIT), *MAKAUT*, Kolkata, India

2016 - 2020 **Bachelor of Technology** (2016-2020 || 861 hours)
Department of Civil Engineering || Department of Computer Science Engineering || Department of Electronics & Communications Engineering || Department of Electrical Engineering || Department of Mecanical Engineering

- **Graph theory & algorithms** (Course + Tutorials || 84 hours)
- **Linear algebra** (Course + Tutorials || 120 hours)
- **Abstract algebra** (Course + Tutorials || 75 hours)
- **Probability** (Course + Tutorials || 132 hours)
- **Real analysis** (Course + Tutorials || 66 hours)
- **Complex analysis** (Course + Tutorials || 127 hours)
- **Numerical methods** (Course + Tutorials + Practicals || 143 hours)
- **Differential calculus** (Course + Tutorials || 134 hours)
- **Integral transforms** (Course + Tutorials || 64 hours)

Government College of Engineering & Ceramic Technology, *MAKAUT*, Kolkata, India

2015 - 2016 **Bachelor of Technology** (2015-2016 || 77 hours)
Department de Ceralic Technologie || Department de Computer Science & Information Technologies Engineering

- **Probability** (Course + Tutorials || 23 hours)
- **Integral transforms** (Course + Tutorials || 17 hours)
- **Fourier analysis** (Course + Tutorials || 17 hours)
- **Differential equations** (Course + Tutorials || 20 hours)

Academic References

- **Florent Foucaud**, *LIMOS, Université Clermont Auvergne*, France
  florent.foucaud@uca.fr
- **Annegret Wagler**, *LIMOS, Université Clermont Auvergne*, France
  annegret.wagler@limos.fr
- **Michael Henning**, *Department of Mathematics and Applied Mathematics, University of Johannesburg*, South Africa
  mahenning@uj.ac.za
- **Sagnik Sen**, *Department of Mathematics, Indian Institute of Technology Dharwad, Karnataka*, India
  sagnik@iitdh.ac.in